VP02 — Architecture and logical decomposition

What parts exist and how are they organized?

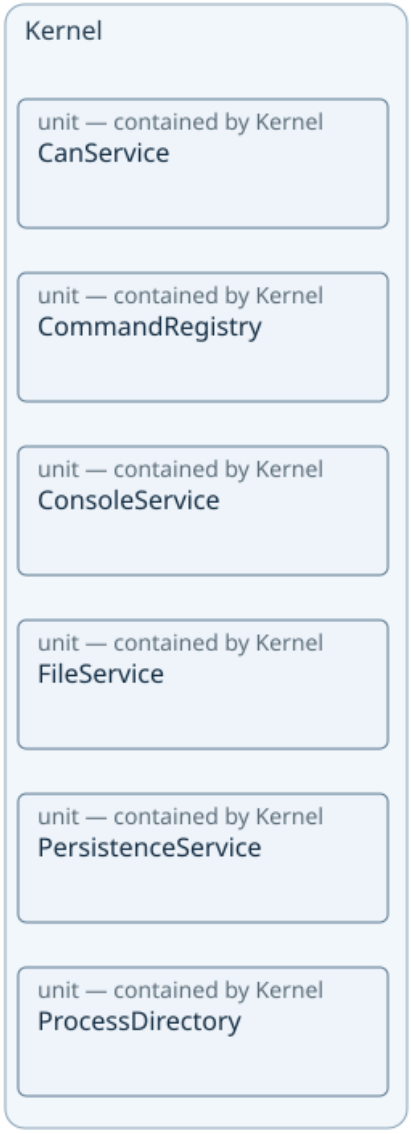
Container/Unit and contains; library structure is not deployment allocation.

Operating mode: All declared modes · NativeCpp · SmallC

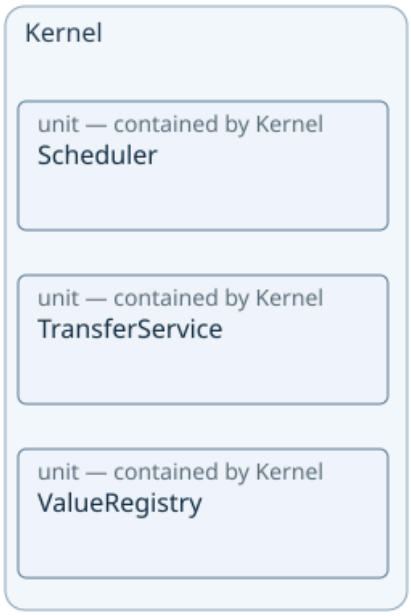
Viewpoint overview

Kernel — participants and channels

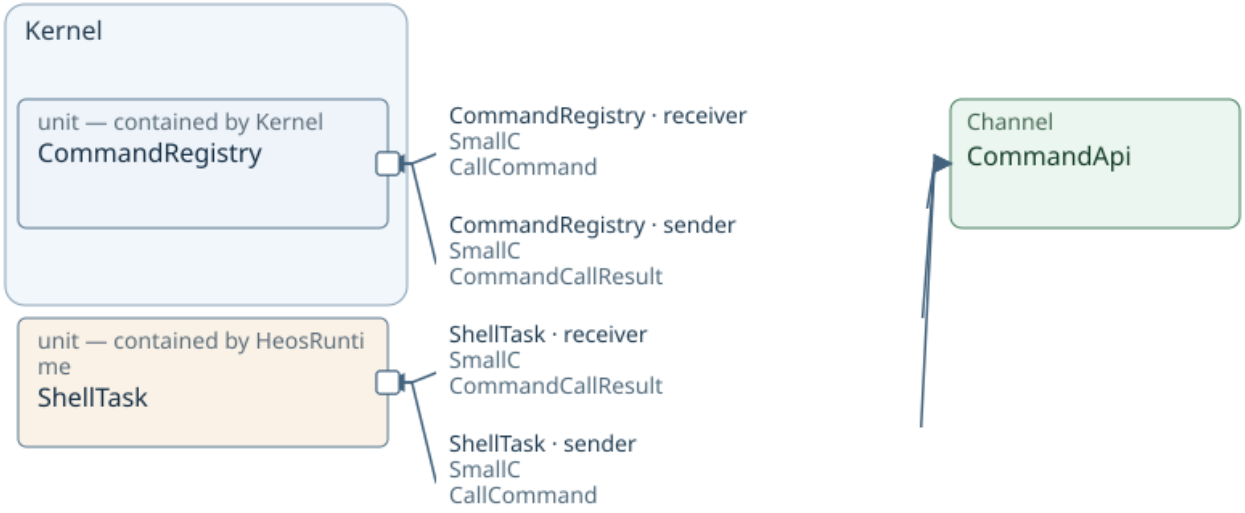
Containment — 1–6 of 9 objects



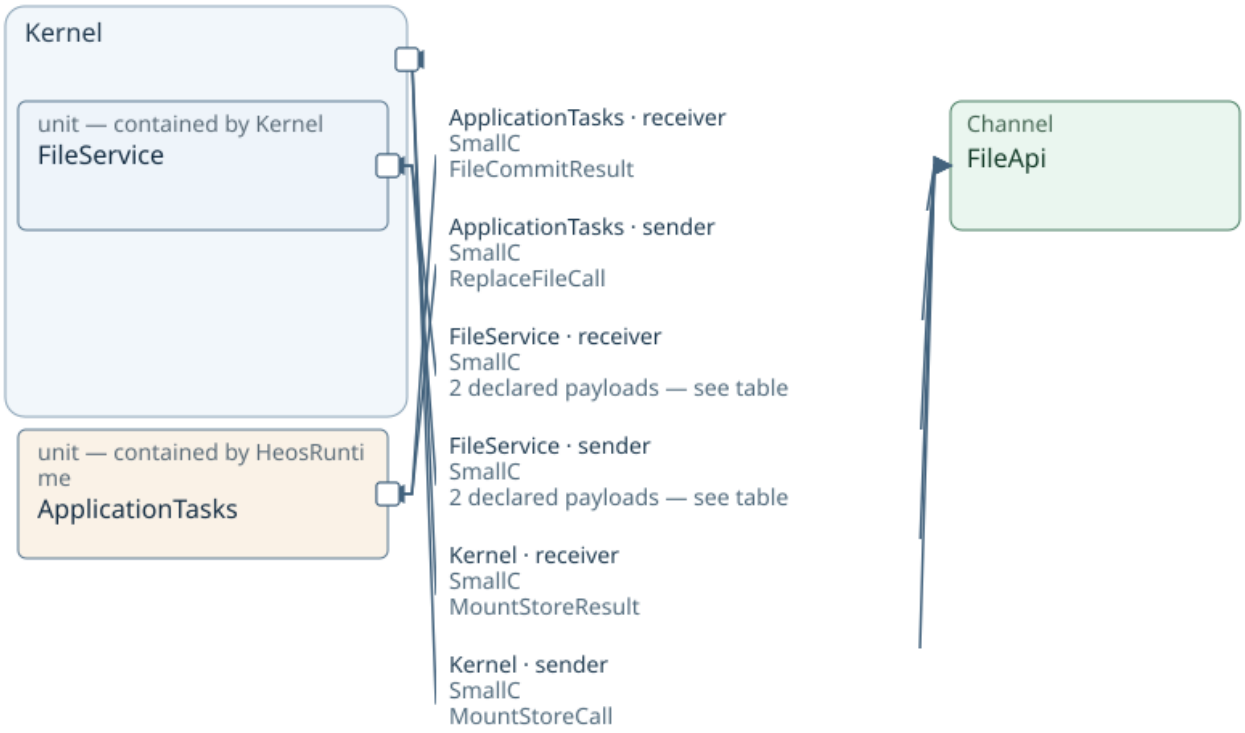
Containment — 7–9 of 9 objects



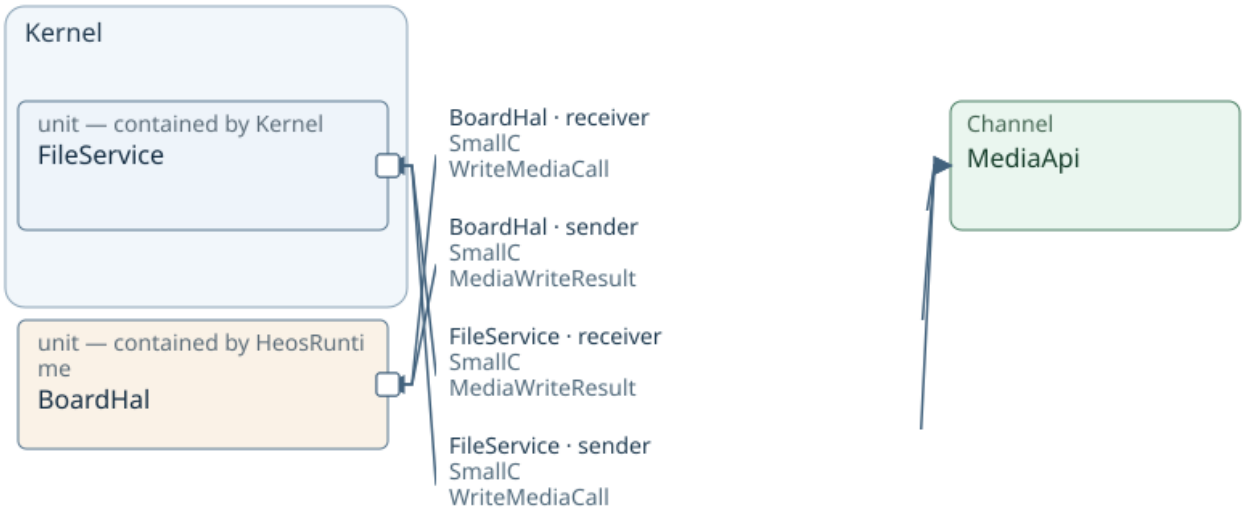
CommandApi — mode SmallC



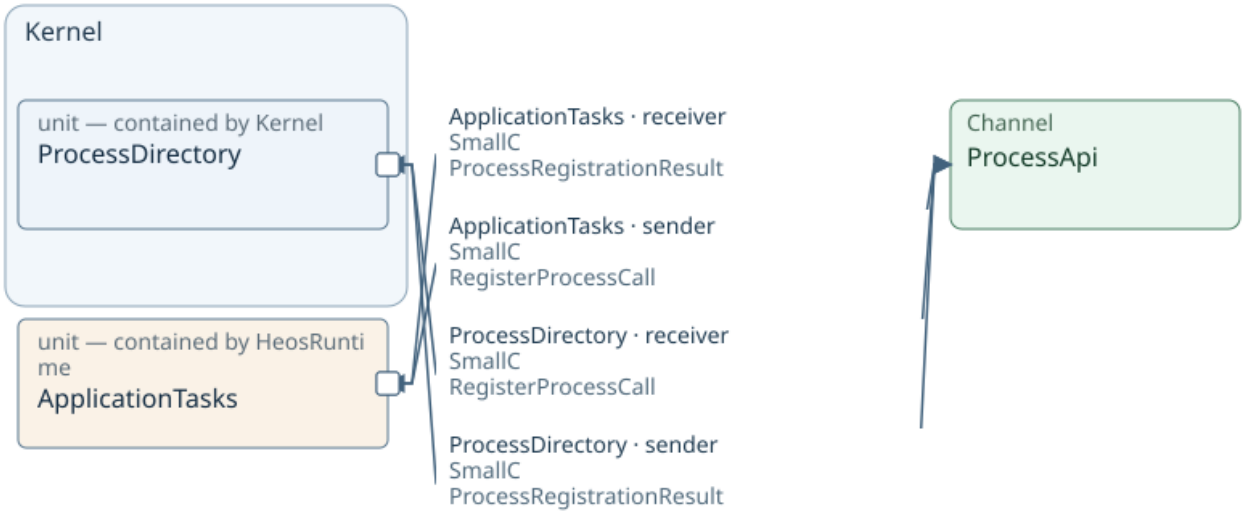
FileApi — mode SmallC



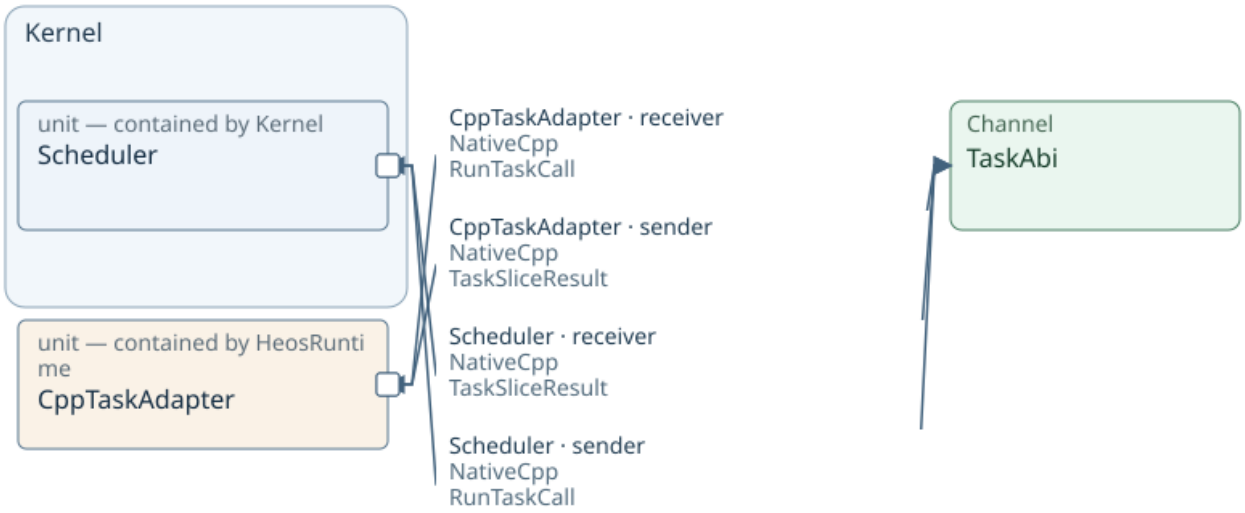
MediaApi — mode SmallC



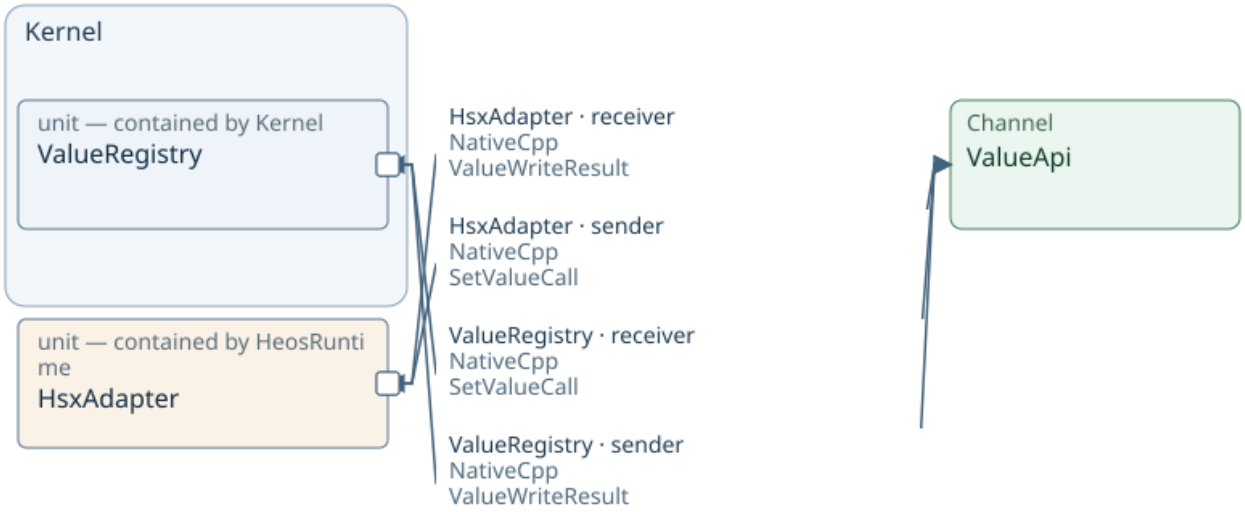
ProcessApi — mode SmallC



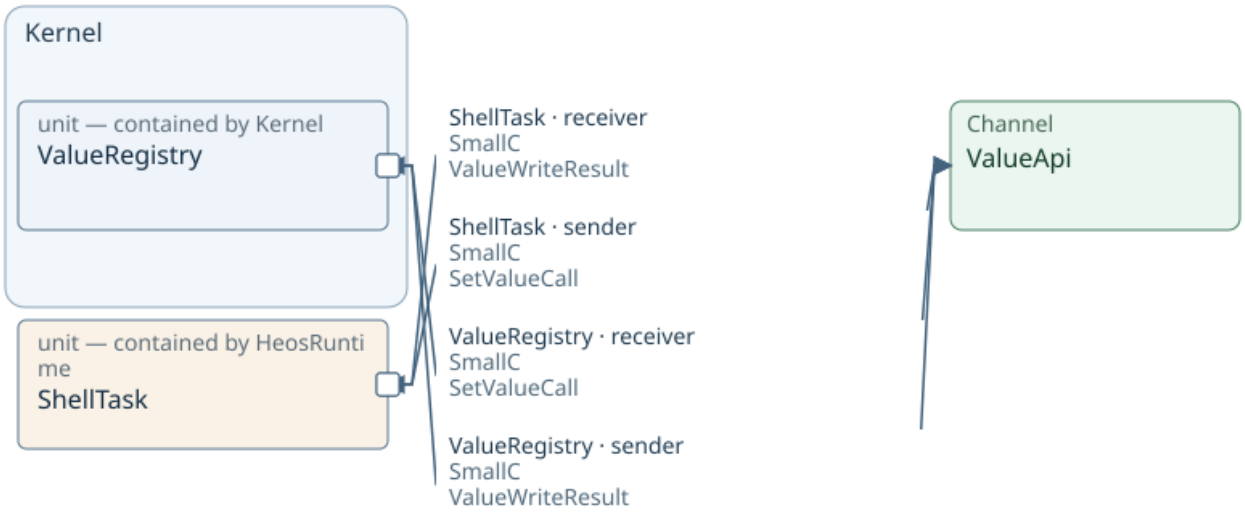
TaskAbi — mode NativeCpp



ValueApi — mode NativeCpp



ValueApi — mode SmallC



Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: CanService · CommandRegistry · ConsoleService · FileService · PersistenceService · ProcessDirectory · Scheduler · TransferService · ValueRegistry · ApplicationTasks · BoardHal · CppTaskAdapter · HsxAdapter · ShellTask

CommandApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
CommandRegistry	receiver	CallCommand	Not applicable	SmallC
CommandRegistry	sender	CommandCallResult	Not applicable	SmallC
ShellTask	receiver	CommandCallResult	Not applicable	SmallC

ShellTask	sender	CallCommand	Not applicable	SmallC

FileApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	FileCommitResult	Not applicable	SmallC
ApplicationTasks	sender	ReplaceFileCall	Not applicable	SmallC
FileService	receiver	MountStoreCall	Not applicable	SmallC
FileService	receiver	ReplaceFileCall	Not applicable	SmallC
FileService	sender	FileCommitResult	Not applicable	SmallC
FileService	sender	MountStoreResult	Not applicable	SmallC
Kernel	receiver	MountStoreResult	Not applicable	SmallC
Kernel	sender	MountStoreCall	Not applicable	SmallC

MediaApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
BoardHal	receiver	WriteMediaCall	Not applicable	SmallC
BoardHal	sender	MediaWriteResult	Not applicable	SmallC
FileService	receiver	MediaWriteResult	Not applicable	SmallC
FileService	sender	WriteMediaCall	Not applicable	SmallC

ProcessApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
ApplicationTasks	receiver	ProcessRegistrationResult	Not applicable	SmallC

ApplicationTasks	sender	RegisterProcessCall	Not applicable	SmallC
ProcessDirectory	receiver	RegisterProcessCall	Not applicable	SmallC
ProcessDirectory	sender	ProcessRegistrationResult	Not applicable	SmallC

TaskAbi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
CppTaskAdapter	receiver	RunTaskCall	Not applicable	NativeCpp
CppTaskAdapter	sender	TaskSliceResult	Not applicable	NativeCpp
Scheduler	receiver	TaskSliceResult	Not applicable	NativeCpp
Scheduler	sender	RunTaskCall	Not applicable	NativeCpp

ValueApi payloads

Participant	Role	Message / Datagram	Dataset origin	Mode
HsxAdapter	receiver	ValueWriteResult	Not applicable	NativeCpp
HsxAdapter	sender	SetValueCall	Not applicable	NativeCpp
ShellTask	receiver	ValueWriteResult	Not applicable	SmallC
ShellTask	sender	SetValueCall	Not applicable	SmallC
ValueRegistry	receiver	SetValueCall	Not applicable	NativeCpp
ValueRegistry	receiver	SetValueCall	Not applicable	SmallC
ValueRegistry	sender	ValueWriteResult	Not applicable	NativeCpp
ValueRegistry	sender	ValueWriteResult	Not applicable	SmallC

Kernel

unit declaration: Kernel — Containers/Runtime/Runtime.design, line 31.

Declared relationships

- ComposeKernel allocated-to HeosRuntime in mode NativeCpp. — Containers/Runtime/Runtime.design, line 107
- ComposeKernel allocated-to HeosRuntime in mode SmallC. — Containers/Runtime/Runtime.design, line 108
- FreezeRegistration allocated-to HeosRuntime in mode NativeCpp. — Containers/Runtime/Runtime.design, line 140
- FreezeRegistration allocated-to HeosRuntime in mode SmallC. — Containers/Runtime/Runtime.design, line 141
- HeosRuntime contains Kernel. — Containers/Runtime/Runtime.design, line 148
- Kernel contains CanService. — Containers/Runtime/Runtime.design, line 162
- Kernel contains CommandRegistry. — Containers/Runtime/Runtime.design, line 163
- Kernel contains ConsoleService. — Containers/Runtime/Runtime.design, line 164
- Kernel contains FileService. — Containers/Runtime/Runtime.design, line 165
- Kernel contains PersistenceService. — Containers/Runtime/Runtime.design, line 166
- Kernel contains ProcessDirectory. — Containers/Runtime/Runtime.design, line 167
- Kernel contains Scheduler. — Containers/Runtime/Runtime.design, line 168
- Kernel contains TransferService. — Containers/Runtime/Runtime.design, line 169
- Kernel contains ValueRegistry. — Containers/Runtime/Runtime.design, line 170
- Kernel owns ComposeKernel. — Containers/Runtime/Runtime.design, line 171

- Kernel owns FreezeRegistration. — Containers/Runtime/Runtime.design, line 172
- Kernel owns ManageLifecycle. — Containers/Runtime/Runtime.design, line 173
- Kernel owns ReportHealth. — Containers/Runtime/Runtime.design, line 174
- Kernel uses FileApi as receiver of MountStoreResult in mode SmallC. — Scenarios/Services.design, line 65
- Kernel uses FileApi as sender of MountStoreCall in mode SmallC. — Scenarios/Services.design, line 66
- ManageLifecycle allocated-to HeosRuntime in mode NativeCpp. — Containers/Runtime/Runtime.design, line 183
- ManageLifecycle allocated-to HeosRuntime in mode SmallC. — Containers/Runtime/Runtime.design, line 184
- ReportHealth allocated-to HeosRuntime in mode NativeCpp. — Containers/Runtime/Runtime.design, line 242
- ReportHealth allocated-to HeosRuntime in mode SmallC. — Containers/Runtime/Runtime.design, line 243

Related objects in this viewpoint

CanService, CommandRegistry, ComposeKernel, ConsoleService, FileApi, FileService, FreezeRegistration, HeosRuntime, ManageLifecycle, MountStoreCall, MountStoreResult, NativeCpp, PersistenceService, ProcessDirectory, ReportHealth, Scheduler, SmallC, TransferService, ValueRegistry

Interpretation and unspecified behavior

Contains describes logical structure. Deployment, scheduling, runtime calls, and unnamed internal units are not inferred. Channel endpoints refer to actual declared participants.